

The Islamic University of Gaza
Faculty of Information Technology
Cloud and Distributed Systems (SICT 4313)

Cloud-Based Distributed Data Processing Service

You are required to design and implement a cloud-based data processing program/service that enables users to upload datasets and perform distributed and parallel machine learning analytics using Apache Spark or PySpark. The program must run in a cloud development platform and deliver results from uploaded data.

- The user uploads dataset (CSV, JSON, Text, PDF, etc.), select processing options, and view/download processed results.
- The program receives and validate uploads, stores files/datasets in cloud storage, starts Spark job for processing, and monitors job status and return results.
- With Spark processing, the program must load data from cloud storage and perform:
 - Descriptive statistics of your choice (such as data sizes, number of rows, number of columns, data types, min/max/mean counts, unique value counts, null/missing value percentages, etc.). You have to display and store at least 4 of these statistics.
 - Machine learning jobs of your choice depending on the type of your uploaded data (such as train a regression model, run KMeans clustering, generate frequent itemsets (FPGrowth), time-series aggregation (daily/weekly/monthly summaries), etc. Perform at least 4 machine learning jobs. Use Spark MLlib and display results on the screen and write results to cloud storage.
- When performing machine learning jobs, run the same Spark job on 1, 2, 4, and 8 machines. Record execution times, compute speedup and efficiency. Include a results table and performance discussion in the report. Based on these results, include a discussion of scalability. To work correctly, these machine learning jobs need large data sizes (which can be found, e.g., on UCI Machine Learning Repository.)

Note:

- You can use any programming language, but preferably Python with pySpark.
- You can use one or more cloud platforms of your choice at the same time, e.g., AWS, GCP, Azure, Databricks, etc.
- The program must have a user friendly interface.

- You must upload the source code of the project to GitHub repository. The source code should be well documented so anyone can read, understand and follow it.
- Write a report (using the attached template) describing, explaining, and discussing your development as well as the platform and how you used it.
- Record a video demonstration on how to use the program (5-7 minutes).
- Include in the report the link to the program on the cloud and how to use it.
- Include in the report the link to the GitHub repository of the program.
- Include in the report the link to the video demonstration.
- You must work in a group of 2-3 students.

Important note: Any use of AI tools to do the program or write/generate the report will result into ZERO in the project and therefore FAILURE in the course.

Best wishes
Dr. Rebhi S. Baraka